

ASSESSMENT : CompTia A+ - Troubleshooting and Helpdesk

Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI

Use an AI tool of your choice (ChatGPT, Claude, Copilot, etc.) to help you produce a step-by-step troubleshooting checklist that

Answer:

AI Tool Used: ChatGPT

1. Exact Prompt Given to the AI

Create a step-by-step troubleshooting checklist for a junior technician to diagnose a desktop PC that fails to boot immediately after assembly.

The checklist should:

- 1.** Follow the correct troubleshooting order from basic checks to component-level checks.
- 2.** Cover at least the motherboard, RAM, storage, and PSU as possible failure points.
- 3.** Include ESD precautions and physical connection checks before testing components.
- 4.** Be presented as a clear checklist or simple decision flow that a technician can follow on-site.
- 5.** Apply the checklist to the ABC Solutions scenario from Section C and explain which troubleshooting step could have identified the reported boot or hardware-detection issue.

2. AI's Original Troubleshooting Checklist

Desktop PC Boot-Failure Troubleshooting Checklist

1. Safety and ESD (Electrostatic Discharge) Check

- Shut down the system and disconnect AC power.
- Use proper ESD precautions before touching internal components.
- Check for loose screws, damaged components, or incorrectly connected cables.

2. Check PSU and Power Connections

- Confirm that the PSU switch is ON.
- Check the motherboard 24-pin ATX connector.
- Check the CPU/EPS power connector.
- Check GPU power connectors if required.
- Confirm that the PSU has sufficient wattage.

3. Check Motherboard

- Check motherboard diagnostic LEDs or beep codes.
- Confirm that the CPU is compatible with the motherboard.
- Check whether the required BIOS/UEFI version supports the CPU.
- Inspect the motherboard for visible damage.

4. Check RAM

- Make sure RAM modules are properly seated.
- Reseat the RAM if necessary.
- Test one RAM module at a time.
- Try another compatible RAM slot if available.

5. Check Storage

- Confirm that the M.2 SSD or SATA drive is properly connected.
- Check whether the storage device is detected in BIOS/UEFI.
- Verify that the correct boot device is selected.

6. Check GPU and Display

- Reseat the GPU if installed.
- Check GPU power connections.
- Confirm that the monitor cable is connected correctly.
- Test the system using integrated graphics if supported.

7. Perform BIOS/POST Verification

- Enter BIOS/UEFI.
- Check CPU, RAM, and storage detection.
- Check for hardware error messages.
- Confirm that POST completes successfully.

8. Isolate the Fault

- Disconnect unnecessary peripherals and components.
- Boot the system with minimum required hardware.
- Add components back one at a time to identify the faulty component.

STEP 2 — TEST & DEBUG (WITHOUT AI)

Missing Step Identified

After reviewing the AI-generated checklist, I found that an explicit **motherboard mounting and short-circuit check** was missing.

Corrected Step

Check Motherboard Mounting and Short Circuits

- Verify that the motherboard is installed only on the correct case standoffs.
- Make sure there are no extra or incorrectly positioned standoffs underneath the motherboard.
- Check for loose screws or metal objects that could cause a short circuit.
- If a short circuit is suspected, remove the motherboard from the case and perform a minimum-hardware POST test outside the case on a non-conductive surface.

Why I Made This Change:

I added this step because an incorrectly positioned motherboard standoff or loose metal object can cause a short circuit and prevent the computer from powering on or completing POST. This check is especially important in the ABC Solutions scenario because the problem occurs immediately after assembly.

Why it was needed:

The ABC Solutions scenario involves newly assembled systems that fail to boot or detect hardware. Incorrect motherboard mounting can directly cause these symptoms, so checking the standoffs and possible short circuits should be part of the physical assembly diagnosis.